

# PS7: AI-Enabled Detection of Exoplanets from TESS Light Curves

ISRO Bharatiya Antariksh Hackathon 2026 | Generated: July 01, 2026

## 1. Problem statement

Develop an AI pipeline to automatically detect exoplanet transit signals in noisy TESS light curves. The pipeline must find periodic brightness dips, classify the signal type (planet transit, eclipsing binary, background blend, or noise), and estimate the orbital period, transit depth, and duration.

## 2. Pipeline overview

- Data acquisition: TESS 2-min cadence light curves downloaded from NASA MAST via lightcurve
- Preprocessing: NaN removal, sigma-clipping (4 sigma), median-filter and Savitzky-Golay detrending compared, normalization
- Transit detection: Box Least Squares (BLS) periodogram scanning 10,000 trial periods (0.5-20 days)
- Cross-validation: Lomb-Scargle periodogram run independently to confirm BLS period detection
- Vetting: Four automated tests — SNR threshold, secondary eclipse check, odd/even depth check, dip symmetry
- Crowding check: TIC catalog queried for bright neighboring stars within 21 arcsec aperture
- Feature engineering: 23 features from BLS output, flux statistics, dip shape, and frequency domain
- Classification: XGBoost (300 estimators, balanced class weights) trained on ISRO-provided labeled dataset
- Parameter estimation: Orbital period (days), transit depth (ppm), transit duration (hours)
- Visualization: Streamlit dashboard with dark space theme and Plotly interactive charts

## 3. Dataset

Primary data: TESS Sector 1 (July-August 2018), 2-minute cadence photometry from the SPOC pipeline. Each light curve contains approximately 19,440 data points over 27 days. Training labels provided by ISRO covering four signal classes: planet, eclipsing binary, blend, and other.

## 4. ML features used

| Feature                 | Description                         | Why it matters             |
|-------------------------|-------------------------------------|----------------------------|
| best_period_days        | BLS best period in days             | Primary identifier         |
| transit_depth_ppm       | Depth of dip in parts per million   | Planet vs EB discriminator |
| transit_duration_hours  | Duration of transit in hours        | Orbital geometry           |
| snr                     | Signal-to-noise ratio of BLS peak   | Detection confidence       |
| secondary_eclipse_depth | Depth at phase 0.5 in ppm           | EB flag: secondary exists  |
| odd_even_ratio          | Ratio of alternating transit depths | EB flag: unequal dips      |
| dip_symmetry            | Left vs right transit depth ratio   | Real transit vs glitch     |
| dip_area                | Integrated area below baseline      | Total blocked light        |

|                            |                               |                              |
|----------------------------|-------------------------------|------------------------------|
| autocorr_peak              | Autocorrelation function peak | Periodicity strength         |
| fft_dominant_freq          | Dominant frequency from FFT   | Periodic signal confirmation |
| flux_std / skew / kurtosis | Light curve shape statistics  | Noise vs signal              |

## Page 2: Results and Model Performance

Classification accuracy and candidate catalog

### 5. Transit candidates detected

The BLS search identified candidates above SNR threshold. After ML classification, planet candidates were flagged for further analysis. Sample results shown below.

| TIC ID    | Period (d) | Depth (ppm) | Duration (hrs) | SNR | Symmetry | Crowding |
|-----------|------------|-------------|----------------|-----|----------|----------|
| 150428135 | 14.8224    | 2192        | 1.20           | 7.8 | 0.00     | —        |

## Page 3: Validation and Uncertainty

Pipeline tested on NASA-confirmed TESS exoplanets

### 6. Validation on known exoplanets

The pipeline was tested on confirmed TESS exoplanets from the NASA Exoplanet Archive. Detected periods are compared to published values. Error less than 2 percent counts as PASS.

| Planet     | TIC ID    | Known period (d) | Detected (d) | Error % | Status |
|------------|-----------|------------------|--------------|---------|--------|
| WASP-126 b | 25155310  | 3.2886           | 3.2887       | 0.0029  | PASS   |
| TOI-700 d  | 150428135 | 37.4228          | 14.8224      | 60.392  | FAIL   |

### 7. Uncertainty estimation

Period uncertainties are estimated by bootstrap resampling: 100 BLS runs on flux-perturbed versions of each light curve, reporting the standard deviation of best-period estimates. Depth uncertainty uses the BLS fit residuals. SNR and the independent Lomb-Scargle cross-check provide additional confidence metrics.

### 8. Tools and libraries

- lightkurve 2.4.2 — TESS light curve download and preprocessing
- astropy BoxLeastSquares and LombScargle — transit detection algorithms
- astroquery — TIC catalog crowding checks
- XGBoost 2.0.3 — gradient-boosted tree classifier with balanced class weights
- scikit-learn — preprocessing, cross-validation, metrics
- Streamlit 1.31.0 — interactive web dashboard

- Plotly 5.18.0 — interactive charts
  - ReportLab 4.0.8 — this PDF report
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## 9. Assumptions

- TESS SPOC pipeline systematics are adequately removed by flatten window of 401 points
- BLS period range 0.5 to 20 days covers most hot Jupiters and warm Neptunes
- SNR greater than 6 is sufficient threshold for transit candidates in 27-day baselines
- Odd/even ratio deviation greater than 30 percent reliably flags eclipsing binaries
- Secondary eclipse greater than 20 percent of primary depth indicates stellar companion
- Bright neighbor stars within 21 arcsec and 3 magnitudes of target indicate crowding risk